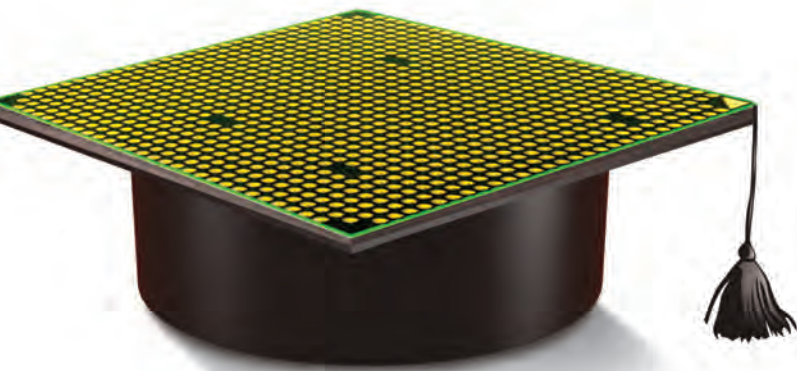




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EDA stands for Electronic Design Automation, and denotes the design tools used for designing electronic systems such as integrated circuits



Be a Chip off the old block

All that you ever wanted to know about a Career in VLSI and Chip Design

Siddharth Parwatay

Chip designing is a challenging job with constant innovation in process technology. In addition to its dynamic nature, there are challenges that need to be overcome as products become increasingly complex. At its base lies Electronics Engineering and Computer Science and like others, this industry too, is not without its lore and legend.

The Indian story of VLSI began in 1985, when Texas Instruments set up the first design centre at Bangalore, thereby ushering in an era of widespread chip design and VLSI. Today, there are several companies in the field including AMD, Intel and NVIDIA, apart from companies engaged in ASIC (Application Specific Integrated Circuit) designs such as KPIT and Mindtree Consulting. VLSI design service companies brought in \$927 million last year according to the ISA (Indian Semiconductor Association), from all three segments — analogue, digital and mixed signal. However, the industry is expected to shift to mixed signal in the coming years. Also, despite the difference in salaries slowly narrowing, having designs produced in India still presents a huge comparative advantage; so the future is bright.

The Indian link to VLSI

Processors and chips developed today contain hundreds of millions of transistors; hence, the term VLSI (very large scale integration). NVIDIA's Indian operation has 780 employees across three sites — Bangalore, Pune and Hyderabad. According to Amit Verma, Director of Human Resources, NVIDIA (India), "In these centres, the focus is on end-to-end ASIC product development encompassing architecture, RTL design, verification, timing, physical design, system engineering (including board bring up and debugging) and quality assurance for chipset and mobile products. NVIDIA's Indian engineers are part of the company's worldwide team and their contribution is vital to the business as a whole."

On the other hand, AMD has two centres in India, where AMD teams usually work on one or two products at a time. This involves a fair bit of interaction with the teams abroad. "Shanghai was, for example, a joint effort between Boston and India", said Karthik Muthusamy, General Manager, India Design Centre at AMD India Engineering Centre

Basic requirements

Passion and having an early understanding of this field is essential. Aspirants will work with sophisticated design tools, chip architectures and test methods. So, apart from technical expertise, additional skill sets are also important. "Key areas that we look for

in our engineers would be strong analytical skills, good problem solving skills, a willingness to work with teams as collaboration is a part of the job profile, and so a lot of communication skills," said Rohit Vidwans, Director — Digital Enterprise Group (DEG), Intel India. Specific to chip design activities, companies look for engineers in Electrical and Computer Science, even Chemical engineers or someone with a strong background in Physics. As for degree certification, companies hire all the way down from a Bachelors degree to a Masters in Science degree, as well as PhD holders; thus, spanning the entire spectrum of educational degrees.

Technical requirements

A strong hardware engineering background with specialisation in VLSI design is a must. Additionally, knowledge of digital systems and VLSI design, computer architecture, computer arithmetic, C/C++ programming languages, CMOS transistors and circuits is necessary. In VLSI design, tools play a major role. An integral part of an engineer's work deals with using the latest industry tools from Synopsys, Cadence and other EDA vendors. An understanding of hardware languages such as Verilog is also important. Other languages such as Perl and C come in handy too. Processor description languages such as nML and Lisa will be an added plus if you are thinking of a career specialisation in microprocessor design.

"Apart from architecture and HDLs, chip designing in today's world will need in-depth knowledge of even transaction level modeling using SystemC, knowledge of electronic design automation tools, and some understanding of the backend timing / manufacturing issues. Knowledge of simulation tools and interactions with memory / cache design will also be needed," says Sandeep Kumar a consultant architect at Coware India Pvt Ltd. Sandeep has sketched a successful career in this field, now spanning over 17 years of experience.

Recruitment

Companies are in search of highly motivated individuals with strong analytical and communication skills. Apart from educational degrees, companies are also interested in the fundamentals. "There are people who understand how to push the tools, but when we talk to candidates we try to see if they are familiar with fundamentals. For example, if we're working on a GCC component we would want to know where you put this particular switch. Why here? Why do you use it? OK, you know how to use ASIC tools, but do you know why you did certain things that way? Ability to understand and pick up is important," said Karthik from AMD. Arriving at the answer is as important as getting to the answer itself. So you can expect these kinds of questions in the interview rounds. Most companies have tests for analytical ability.

Talent is sourced by tying up with premier schools in the